

SAISANJAY R

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EDUCATION

Kumaraguru College of Technology, Coimbatore

Expected May 2028

B.Tech in Artificial Intelligence & Data Science | CGPA: 8.25 / 10.00

TECHNICAL SKILLS

- **Languages:** Python (Advanced), C, C++, MySQL
- **AI/ML & Frameworks:** JAX, PyTorch, TensorFlow, Keras, LangChain, LlamaIndex, Weights & Biases (W&B), TensorBoard
- **Databases & Tools:** MongoDB (Beanie ODM), ChromaDB (Vector DB), Docker, Docker, Git, GitHub

PROJECTS

PRRegression Audit — Multi-Agent RL Code Reviewer

Apr 2026

Technologies: Python, FastAPI, GRPO Fine-tuning, Reinforcement Learning, Multi-Agent Systems, Docker, W&B

- Engineered an autonomous 4-agent pipeline (Safety, Defect, Router, Comment) for PR regression detection.
- Prevented reward hacking by deploying RLVR (Verifiable Rewards) for deterministic, zero-LLM programmatic grading.
- Achieved **2.1× performance gain** on Medium tasks (reward 0.0382 → 0.5300) fine-tuning Qwen2.5-1.5B via GRPO.
- Shipped a live W&B tracking dashboard on HuggingFace Spaces backed by 12+ async FastAPI endpoints.

Smart Document Hub — Multilingual Document Intelligence

Jun 2026 — In Progress

Technologies: FastAPI, Next.js, LangChain, LlamaIndex, ChromaDB, MongoDB (Beanie ODM), Celery, Redis, spaCy, EasyOCR

- Architected an async FastAPI + Beanie ODM backend designed to maintain document Q&A latency **under 2 seconds**.
- Built a Celery + Redis background task pipeline to isolate heavy AI processing from the core request-response cycle.
- Implemented a hybrid deadline-extraction engine using local spaCy NER and LangChain/Gemini classification.
- Constructed a document knowledge graph using vector-space cosine similarity on ChromaDB embeddings, visualized via D3.js.

CardioPredict — Low-Level MLP Regression Model

May 2026

Technologies: JAX, XLA Acceleration, Functional Programming

- Bypassed high-level wrappers to build an MLP from scratch using pure functional programming math.
- Leveraged JAX primitives (vmap for vectorization, grad for reverse-mode autodiff) to optimize XLA training.
- Achieved a **Test MSE of 10.22** over 1,000 epochs using strict feature standardization.

EXTRACURRICULAR & ACHIEVEMENTS

- **Algorithmic Consistency:** Solved **100+ DSA problems on LeetCode** focusing heavily on optimization and runtime complexity.
- **Academic Merit:** Mahatma Gandhi Merit Scholarship — Awarded for sustained academic excellence at KCT, 2025–26.
- **Innovation:** 3rd Place out of competitive institutional cohorts, Open Ideathon, Kumaraguru College of Technology.
- **Leadership:** Operations Lead — GDG on Campus; spearheaded backend and pipeline debugging challenges for student bootcamps.